



Week 9 Workshop

Learning to Rank

IFN647

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Workshop Objectives

1. Develop and implement feature selection techniques using probabilistic models (e.g., BM25-based term weighting) to construct a user relevance model (Task 1).
2. Apply the learned relevance model to document ranking, where documents are scored based on the selected features and their associated weights, forming an approximation of a feature distribution (Task 2).
3. Evaluate the effectiveness of the ranking results using appropriate evaluation metrics (Task 3).

Workshop Task 1

Task 1. (Feature selection from training set D)

Design a python program to evaluate terms weights (define function w_5) using the above procedure; and then select top-terms (e.g., their weights are great than the *mean* of the w_5 weights of terms $+ \theta$), where θ is an experimental parameter. $\theta = 3.5$

You can firstly design a function w_5 using the following

Inputs: Training_set folder, Training_benchmark.txt and θ

Output: a dictionary of features with their w_5 weights

Then in the main program, save the features to a text file (Model_w5_R102.dat) with the following outcomes:

(1) Access the training set

- Get parsed documents from the training set

(2) Access the benchmark text file

- Get relevance judgment for each doc id

(3) Calculate w_5 score

(4) Select top features based on threshold

$$w_5(t) = \log \frac{\frac{r(t) + 0.5}{R - r(t) + 0.5}}{\frac{n(t) - r(t) + 0.5}{(N - n(t)) - (R - r(t)) + 0.5}}$$

rapist 106.20000000000002
multipl 39.77987421383647
unemploy 37.85714285714286
site 35.98159509202454
convict 31.774193548387096
featur 30.62130177514793
mistak 22.48603351955307
unknown 20.96685082872928
present 19.48087431693989
grief 18.027027027027028
defend 16.604278074866308
father 15.8064
ground 15.21164021164021
bungl 14.627906976744185
inquiri 14.337634408602149
cabinet 13.848167539267015
debat 13.848167539267015
plus 13.848167539267015
wive 13.848167539267015
bewild 13.848167539267015
earli 13.404968944099378
fate 13.029350104821802
nine 12.991111111111111
...

Workshop Task 2

Task 2. Rank documents in Test_set folder (U) based on *Features* to test the BM25 model.

- (1) Design an algorithm, “**procedure** $BM25Testing(Features, U)$ ”, to calculate BM25 ranking score for all documents in U and return a dictionary of {doc1: rank1, ...}.
- (2) Design a Python function to implement **procedure** $BM25Testing(U, Features)$, and save all documents in a file name “rankBM25.txt” as follows:

```
3827 520.5045275357625
3833 516.1318289439776
4306 492.4045372037915
9703 488.95511782538625
8333 445.94029629025107
7118 408.65233456451114
3828 404.2271828695591
7502 394.9948992699487
9790 386.9413050322148
6498 385.8329976942741
```

$$\text{relevance}(d) = \sum_{t \in T} w(t) \tau(t, d),$$



Act as a long query



Model_w5_R102.dat

Workshop Task 3

Task 3. Read a ranking result file (e.g., “rankBM25.txt”), calculate its Average Precision.

Design a python program to

- (1) Calculate Recall and Precision at rank positions where a relevant document was retrieved.
- (2) Calculate the average precision.
Print out the results as follows:

At position 1, precision= 1.0, recall= 0.034482758620689655
At position 2, precision= 1.0, recall= 0.06896551724137931
At position 3, precision= 1.0, recall= 0.10344827586206896
At position 4, precision= 1.0, recall= 0.13793103448275862
At position 5, precision= 1.0, recall= 0.1724137931034483
At position 6, precision= 1.0, recall= 0.20689655172413793
At position 7, precision= 1.0, recall= 0.2413793103448276
At position 8, precision= 1.0, recall= 0.27586206896551724
At position 9, precision= 1.0, recall= 0.3103448275862069
At position 10, precision= 1.0, recall= 0.3448275862068966
...

---The average precision = 1.0

	Relevant	Non-Relevant
Retrieved	$A \cap B$	$\overline{A \cap B}$
Not Retrieved	$A \cap \overline{B}$	$\overline{A \cap \overline{B}}$

$$Recall = \frac{|A \cap B|}{|A|}$$

$$Precision = \frac{|A \cap B|}{|B|}$$

$$Recall@n = \frac{|A \cap B_n|}{|A|}$$

$$Precision@n = \frac{|A \cap B_n|}{n}$$

where B_n is the set of **top-n** documents in the output of the IR model.